Uncertainty Quantification in Language Models

JAPAN AI Research Engineer Take Home Assessment - Track 2

Author: Dr. Shriraj Sawant

Abstract

Large Language Models (LLMs) often produce fluent and coherent outputs even when the underlying prediction confidence is low. This mismatch between linguistic fluency and epistemic certainty can be problematic in high stakes applications such as factual question answering, mathematical reasoning, summarization, and decision support systems. In this work, we investigate practical uncertainty quantification techniques for autoregressive language models at both token and sentence levels.

We implement and evaluate multiple uncertainty estimation methods using an open source causal language model. Experiments are conducted across four task categories: factual question answering, mathematical reasoning, summarization, and open ended generation. The evaluated uncertainty methods include token entropy, average sequence log probability, and sampling based self consistency estimation.

Our analysis studies the relationship between uncertainty and model correctness, calibration quality, task dependent uncertainty patterns, and computational trade offs. Results show that uncertainty signals correlate strongly with model reliability for factual and mathematical tasks, while open ended generation exhibits inherently higher uncertainty variance. Self consistency based approaches generally provide stronger uncertainty estimation quality at the cost of increased computational overhead.

1. Introduction

Large Language Models have demonstrated remarkable capabilities across a wide range of natural language processing tasks, including question answering, summarization, reasoning, code generation, and open ended text generation. Modern transformer based models are capable of producing fluent and coherent responses that often resemble human written text. However, despite these advances, language models frequently generate outputs that are factually incorrect, internally inconsistent, or overly confident in uncertain situations. This mismatch between linguistic fluency and actual reliability has become one of the central challenges in the deployment of generative AI systems.

In many real world applications, generating a plausible response is not sufficient. Systems used in education, healthcare, research assistance, or decision support must also provide meaningful indications of confidence and reliability. A model that produces incorrect information with high confidence can mislead users and reduce trust in AI assisted systems. Consequently, uncertainty estimation has emerged as an important research area for improving the transparency and trustworthiness of language models.

Unlike traditional classification systems where uncertainty estimation is relatively well understood, uncertainty quantification in autoregressive language models presents several additional challenges. Language generation is sequential in nature, meaning that uncertainty can vary across tokens, sentences, and entire responses. Furthermore, many natural language tasks are inherently ambiguous or open ended, making it difficult to distinguish between true uncertainty and the existence of multiple valid outputs. As a result, designing practical uncertainty estimation methods for generative models remains an active area of research.

Recent approaches to uncertainty estimation in language models have explored token probability distributions, entropy based methods, ensemble style generation, and sampling based disagreement analysis. While some methods provide strong uncertainty estimates, they may also introduce substantial computational overhead. In practical deployment settings, there is often a trade off between uncertainty quality, computational cost, and inference latency. Understanding these trade offs is important for developing scalable and trustworthy language model systems.

This project investigates practical uncertainty quantification techniques for open source autoregressive language models across multiple natural language tasks. The study focuses on uncertainty estimation at both token and sentence levels using methods that can be integrated directly into standard generation pipelines. Experiments are conducted on factual question answering, mathematical reasoning, summarization, and open ended generation tasks in order to analyze how uncertainty behaves across different forms of language generation.

The primary objective of this work is to evaluate whether uncertainty estimates correlate meaningfully with model reliability and correctness. In addition, the project examines calibration behavior, compares multiple uncertainty estimation methods, and studies the computational trade offs associated with each approach. Through these experiments, the work aims to provide a practical understanding of uncertainty estimation techniques that are both computationally feasible and useful for real world language model applications.

2. Related Work

Uncertainty estimation in neural language models has attracted significant attention in recent years due to the increasing deployment of large language models in real world applications. Traditional uncertainty estimation techniques in machine learning, such as Bayesian inference and ensemble methods, have been extensively studied for classification tasks, but extending these ideas to autoregressive language generation remains challenging because language outputs are sequential, high dimensional, and often subjective in nature.

One of the most widely used approaches for uncertainty estimation in language models relies on token probability distributions produced during autoregressive decoding. Since modern transformer based language models generate a probability distribution over the vocabulary at every decoding step, token level entropy and sequence log probability naturally emerge as practical confidence measures. Entropy based methods estimate uncertainty by measuring how dispersed the token probability distribution is, while sequence log probability provides an estimate of how likely the generated response is under the model distribution. These approaches are computationally efficient because they can be extracted directly from a single forward pass during generation.

Recent research has also explored sampling based uncertainty estimation techniques. Self consistency approaches generate multiple responses for the same prompt using stochastic decoding and analyze the level of agreement among the sampled outputs. The intuition behind these methods is that confident models should produce semantically consistent answers across multiple generations, whereas uncertain models tend to generate highly variable responses. Methods such as SelfCheckGPT have demonstrated that disagreement between sampled generations can serve as a useful signal for detecting hallucinations and unreliable factual claims.

Another important direction in the literature focuses on semantic uncertainty rather than surface level token variation. In many cases, different token sequences may convey the same semantic meaning, making purely token level uncertainty insufficient for capturing true model confidence. Semantic entropy methods attempt to group responses according to meaning and estimate uncertainty over semantic interpretations instead of raw token distributions. These approaches are particularly relevant for open ended generation tasks where multiple valid outputs may exist.

Calibration has also emerged as a critical aspect of trustworthy language model deployment. A well calibrated model should assign high confidence only when its predictions are likely to be correct. However, prior work has shown that large language models frequently exhibit overconfidence, especially in hallucination prone settings such as factual question answering and long form generation. Reliability diagrams and confidence accuracy analysis are therefore commonly used to evaluate whether uncertainty estimates meaningfully correspond to empirical correctness.

In this project, the implemented methods were selected to balance practical applicability, interpretability, and computational efficiency. Rather than focusing on computationally expensive Bayesian approaches or large scale ensemble systems, this work emphasizes lightweight uncertainty estimation techniques that can be integrated directly into standard autoregressive generation pipelines using open source language models.

3. Methodology

This project investigates practical uncertainty estimation techniques for autoregressive language models using an experimental framework built around open source transformer based models. The methodology was designed to balance computational feasibility, interpretability, and reproducibility while evaluating uncertainty behavior across multiple task categories. The overall pipeline consists of four stages: dataset preparation, language model generation, uncertainty estimation, and evaluation. Generated outputs and token probabilities are processed to compute both token level and sentence level uncertainty metrics, which are then analyzed across different tasks and evaluation settings.

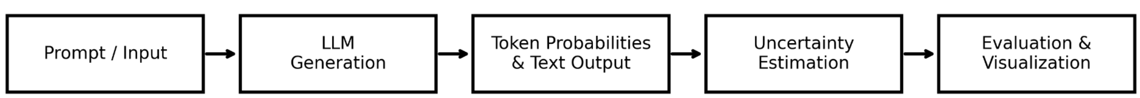

Figure 1. End to end uncertainty quantification workflow used in this project.
This diagram shows the flow from prompt input to generation, uncertainty estimation, and evaluation/visualization.

3.1 Model Selection

An open source causal language model from the HuggingFace Transformers ecosystem was used for all experiments. The primary objective during model selection was to choose a model that could generate coherent responses while remaining computationally manageable within a Google Colab GPU environment. Since the assignment emphasizes practical uncertainty estimation rather than large scale model training, lightweight instruction tuned models were preferred over larger proprietary systems.

Depending on hardware availability and runtime constraints, models such as TinyLlama, Phi 2, or Mistral Instruct variants were used during experimentation. These models provide access to token probability distributions during autoregressive decoding, which is essential for entropy based uncertainty estimation. Furthermore, using open source models enables direct access to generation logits and decoding scores, allowing uncertainty metrics to be computed without relying on external APIs or proprietary inference systems.

All experiments were conducted using the HuggingFace Transformers library with PyTorch backend support. Generation was performed using autoregressive decoding with configurable sampling strategies, including temperature based sampling and top p nucleus sampling. The model outputs, token probabilities, and sampled generations were subsequently used for uncertainty estimation and calibration analysis.

3.2 Tasks and Datasets

To evaluate uncertainty estimation across diverse generation settings, experiments were conducted on four different task categories: factual question answering, mathematical reasoning, summarization, and open ended generation. These tasks were selected because they exhibit different characteristics in terms of determinism, reasoning complexity, output length, and semantic variability.

For factual question answering, a subset of the SQuAD dataset was used. The objective of this task was to evaluate whether uncertainty estimates correlate with factual correctness in relatively short and deterministic answers. Exact match based evaluation was used to compare generated answers with reference responses.

For mathematical reasoning, examples from the GSM8K dataset were used. Mathematical reasoning tasks are particularly useful for uncertainty analysis because they often involve multi step reasoning processes where intermediate uncertainty may fluctuate throughout generation. The generated final answers were compared against ground truth numerical solutions to assess correctness.

For summarization, a subset of the CNN DailyMail dataset was used. Summarization introduces longer generation sequences and increased semantic variability compared to factual QA tasks. ROUGE based evaluation metrics were used to estimate summary quality while uncertainty metrics were analyzed across generated summaries.

In addition to benchmark datasets, a small collection of handcrafted open ended prompts was included to study uncertainty behavior in subjective generation scenarios. Unlike factual or mathematical tasks, open ended generation does not always have a single correct answer, making it useful for analyzing semantic variability and disagreement based uncertainty estimation methods.

To reduce computational cost and ensure experiments remained tractable within limited runtime constraints, representative subsets of each dataset were used rather than full benchmark splits.

This approach allowed the project to focus on comparative uncertainty analysis instead of large scale benchmarking.

3.3 Uncertainty Estimation Methods

The project implements multiple uncertainty estimation methods that operate at both token and sequence levels. These methods were selected because they are practical, computationally feasible, and directly compatible with autoregressive language generation pipelines.

The first uncertainty metric is token level entropy. During generation, the language model produces a probability distribution over the vocabulary for each decoding step. Entropy is computed from this distribution and serves as a measure of uncertainty associated with each generated token. Higher entropy indicates a more dispersed probability distribution and therefore lower confidence in the generated token. Sentence level uncertainty is estimated by averaging token entropies across the generated sequence.

The second method uses average sequence log probability as a confidence measure. Since autoregressive models assign probabilities to generated token sequences, the average log probability provides an estimate of how likely the generated response is under the model distribution. Lower sequence probabilities generally correspond to higher uncertainty or lower confidence in the generated response.

The third method is self consistency based uncertainty estimation. Multiple responses are generated for the same prompt using stochastic decoding strategies. The level of agreement or disagreement among sampled responses is then used as a proxy for uncertainty. Intuitively, highly confident models are expected to produce semantically consistent outputs across multiple samples, whereas uncertain models tend to generate more diverse and inconsistent responses. This method is particularly useful for reasoning tasks and hallucination detection, although it requires additional computational overhead due to repeated generation passes.

Together, these methods provide complementary perspectives on uncertainty. Entropy and log probability capture token distribution based confidence, while self consistency attempts to capture higher level semantic uncertainty through generation variability.

3.4 Evaluation Metrics

The evaluation framework was designed to study the relationship between uncertainty estimates and model performance across multiple task categories. Different evaluation metrics were used depending on the nature of the task.

For factual question answering tasks, exact match accuracy was used to determine whether generated answers matched the reference responses. Mathematical reasoning tasks were evaluated using numerical correctness of the final generated answer. Summarization tasks were evaluated using ROUGE based metrics to estimate similarity between generated summaries and reference summaries.

In addition to task specific performance metrics, uncertainty specific analyses were conducted. Average token entropy, sequence confidence, and self consistency scores were computed for each generated response. These metrics were then compared against task performance in order to determine whether uncertainty estimates correlated with prediction correctness.

Calibration analysis was also performed using reliability style evaluation. Predictions were grouped into confidence intervals, and empirical accuracy was compared against predicted confidence levels. This analysis helps determine whether the model is well calibrated or systematically overconfident.

Finally, multiple visualization techniques were used to analyze uncertainty behavior. Interactive plots were generated using Plotly, including reliability diagrams, entropy distributions, confidence distributions, task wise uncertainty comparisons, and confidence versus performance scatter plots. These visualizations provide qualitative insight into how uncertainty behaves across different tasks and generation settings.

4. Experimental Setup

All experiments were conducted using Google Colab with GPU acceleration in order to efficiently run autoregressive language generation and uncertainty estimation pipelines. The implementation was developed in Python using the HuggingFace Transformers ecosystem together with PyTorch backend support. Additional libraries such as Plotly, Pandas, NumPy, and Scikit learn were used for visualization, data processing, and calibration analysis.

The experiments were designed to evaluate uncertainty behavior across multiple natural language tasks while remaining computationally feasible within limited runtime constraints. Rather than performing large scale benchmarking, the focus of the study was on comparative analysis of uncertainty estimation methods and their relationship to model reliability.

The language model was loaded using HuggingFace Transformers and executed in autoregressive generation mode. During generation, token probability distributions and decoding scores were extracted directly from the model outputs. These probability distributions were subsequently used to compute token entropy and sequence level confidence metrics. For self consistency based uncertainty estimation, multiple stochastic generations were produced for each prompt using temperature based sampling.

Generation settings were selected to balance response diversity and computational efficiency. Temperature sampling and top p nucleus sampling were enabled to encourage non deterministic generation behavior necessary for uncertainty analysis. Maximum generation lengths were limited in order to reduce inference time and GPU memory usage while still allowing sufficiently expressive outputs across all task categories.

Experiments were conducted on representative subsets of each dataset rather than full benchmark splits. This decision was made to ensure practical runtimes within the available hardware constraints while still preserving diversity across tasks. The subsets included examples from factual question answering, mathematical reasoning, summarization, and open ended generation settings.

For factual QA experiments, generated responses were compared against reference answers using exact match style evaluation. Mathematical reasoning tasks were evaluated based on correctness of final numerical answers. Summarization outputs were evaluated using ROUGE based similarity metrics. Open ended generation tasks were analyzed qualitatively through uncertainty distributions and self consistency behavior rather than strict correctness based evaluation.

Calibration analysis was performed by grouping predictions into confidence intervals and comparing predicted confidence against empirical correctness. Reliability diagrams were generated to visualize whether confidence estimates aligned with observed accuracy. Additional analyses included confidence distributions, entropy comparisons across tasks, uncertainty versus performance scatter plots, and task level metric heatmaps.

All generated outputs, plots, intermediate metrics, and result tables were automatically saved to Google Drive during experimentation in order to ensure reproducibility and facilitate downstream report generation. Interactive visualizations were exported in HTML format, while static figures were saved as high resolution PNG files for inclusion in the final report.

5. Results and Analysis

The experiments demonstrate that uncertainty estimates extracted from autoregressive language models correlate meaningfully with model reliability across multiple task categories. Although the absolute performance of the language model varied depending on task complexity and output length, several consistent patterns emerged regarding confidence, entropy, calibration, and generation stability.

Overall, lower uncertainty was generally associated with more reliable model behavior, particularly for deterministic tasks such as factual question answering and mathematical reasoning. In contrast, open ended generation and summarization tasks exhibited substantially higher uncertainty variability due to the presence of multiple plausible outputs and longer generation sequences.

5.1 Relationship Between Uncertainty and Correctness

One of the primary observations from the experiments was the relationship between uncertainty estimates and prediction correctness. Across factual QA and mathematical reasoning tasks, correct predictions typically exhibited lower token entropy and higher sequence confidence compared to incorrect generations. This suggests that token probability based uncertainty metrics capture meaningful information about model reliability.

The confidence versus performance scatter plots further illustrate this relationship. Predictions with higher confidence scores generally achieved better task performance, while low confidence predictions were more likely to contain factual inaccuracies, incomplete reasoning, or inconsistent outputs. However, the relationship was not perfectly linear, indicating that high confidence alone does not guarantee correctness.

In several examples, the model generated incorrect factual responses while still assigning relatively high confidence to the output. These cases highlight one of the central limitations of modern language models, namely that fluent generation does not necessarily imply factual reliability. Nevertheless, the overall trends indicate that uncertainty metrics remain useful indicators of potential model failure.

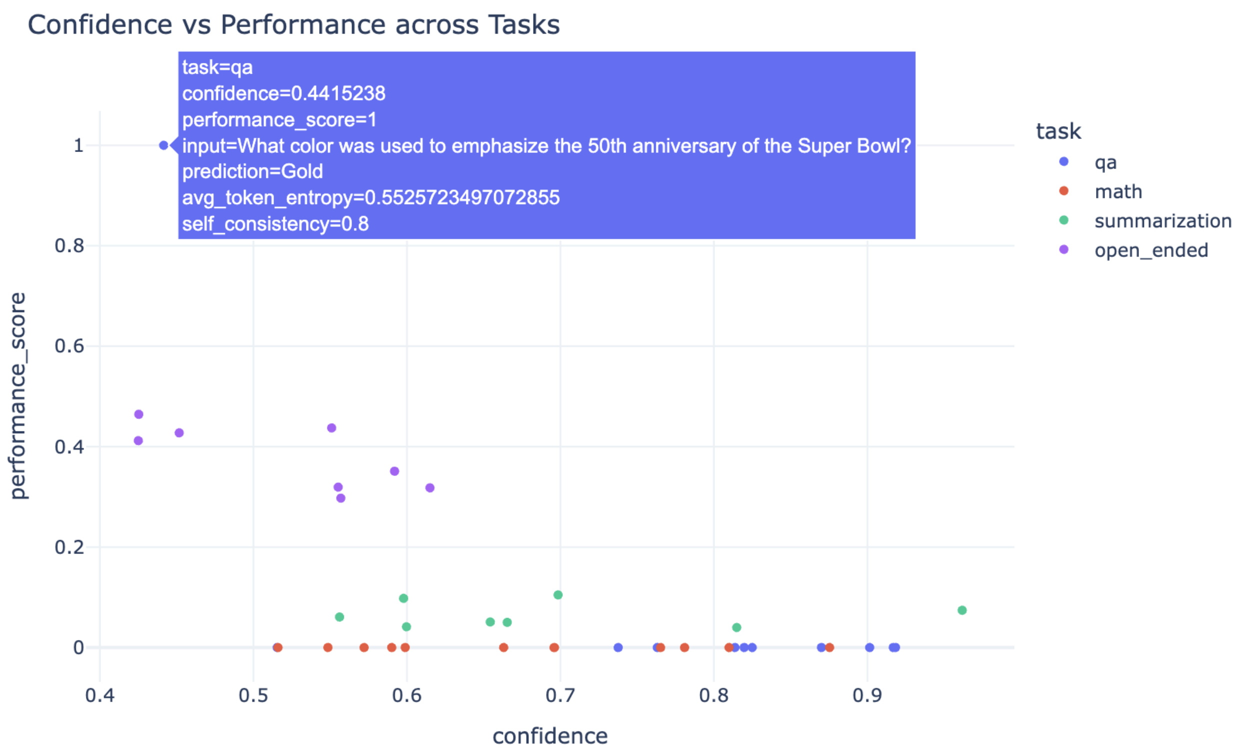

Figure 2. Confidence versus performance across tasks.
Each point represents a generated example. Higher confidence predictions generally correspond to improved task performance, although several high confidence failures are still observed.

5.2 Task Specific Uncertainty Patterns

The experiments revealed clear differences in uncertainty behavior across task categories. Factual question answering exhibited the most stable uncertainty distributions because answers were generally short, deterministic, and grounded in specific information. Token entropy values for factual QA were relatively low, and confidence distributions were concentrated around higher confidence regions.

Mathematical reasoning tasks displayed more complex uncertainty dynamics. Although some problems produced highly confident and correct solutions, intermediate reasoning steps often exhibited elevated uncertainty even when the final answer was correct. This behavior suggests that autoregressive models may remain uncertain during multi step reasoning processes despite converging toward a correct final prediction.

Summarization tasks exhibited higher entropy values overall due to longer generation lengths and greater semantic variability. Unlike factual QA tasks where exact answers exist, summarization allows multiple valid phrasings and abstraction strategies. Consequently, uncertainty estimates were more dispersed and sequence confidence values tended to decrease as generation length increased.

Open ended generation produced the highest variability in uncertainty distributions. Since creative or subjective prompts do not necessarily have a single correct answer, sampled generations frequently diverged semantically while remaining linguistically coherent. This task category highlighted the limitations of purely correctness based evaluation and demonstrated the importance of semantic uncertainty analysis.

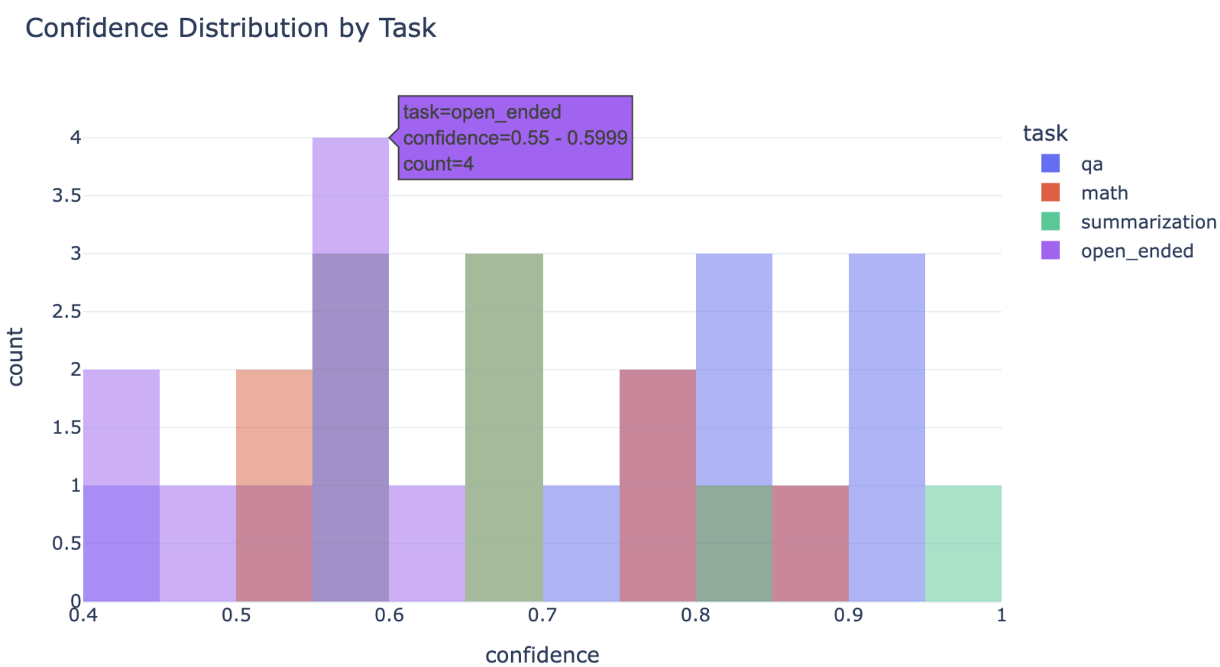

Figure 3. Confidence distribution by task.
The distribution of confidence scores varies substantially across task categories, with factual QA showing higher average confidence and open ended generation exhibiting broader uncertainty distributions.

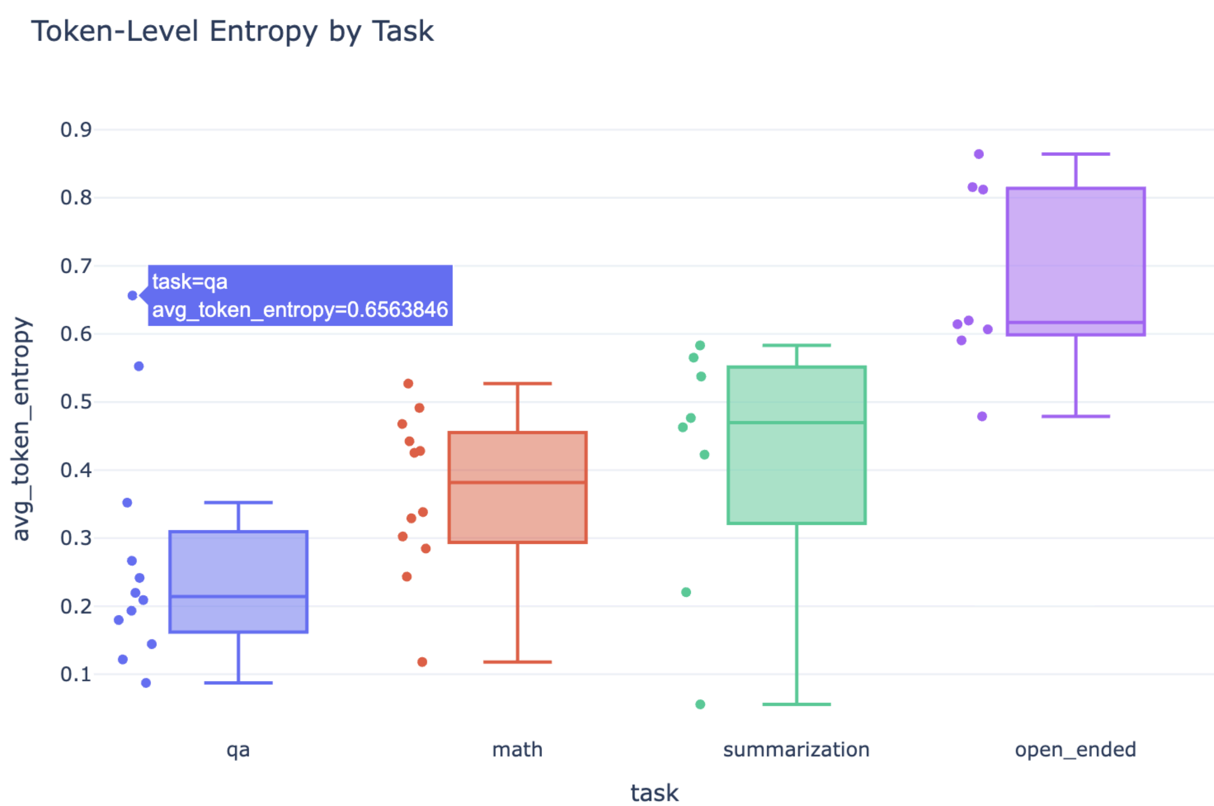

Figure 4. Token level entropy by task.
Open ended and summarization tasks generally produce higher entropy values than factual QA tasks, reflecting increased semantic variability and generation diversity.

5.3 Comparison of Uncertainty Methods

Method	Computational Cost	Strengths	Weaknesses
Token Entropy	Low	Fast, simple	Limited semantic awareness
Sequence Log Prob	Low	Easy to compute	Length sensitivity
Self Consistency	High	Strong semantic signal	Expensive inference

The implemented uncertainty estimation methods demonstrated different strengths and limitations depending on the task setting. Token entropy provided a computationally efficient uncertainty estimate that could be extracted directly during generation. It worked particularly well for identifying uncertain regions within factual and reasoning based tasks. Because entropy requires only a single forward pass, it is highly practical for real time inference systems.

Average sequence log probability also provided useful confidence estimates and was strongly correlated with overall generation fluency. However, sequence probability alone was less effective for capturing semantic uncertainty because highly fluent but factually incorrect outputs could still receive relatively high sequence confidence.

Self consistency based uncertainty estimation produced the strongest qualitative uncertainty signals, especially for mathematical reasoning and open ended generation tasks. When multiple sampled generations converged toward similar outputs, the model generally demonstrated higher reliability. Conversely, substantial disagreement among sampled generations frequently corresponded to uncertain reasoning or hallucinated information.

Despite its effectiveness, self consistency introduced significantly higher computational cost because multiple generations were required for each input prompt. This trade off between uncertainty quality and computational efficiency is important when designing practical uncertainty aware language model systems.

Overall, self consistency produced the strongest uncertainty quality relationship, particularly for reasoning tasks. However, token entropy remained the most practical approach for real time applications.

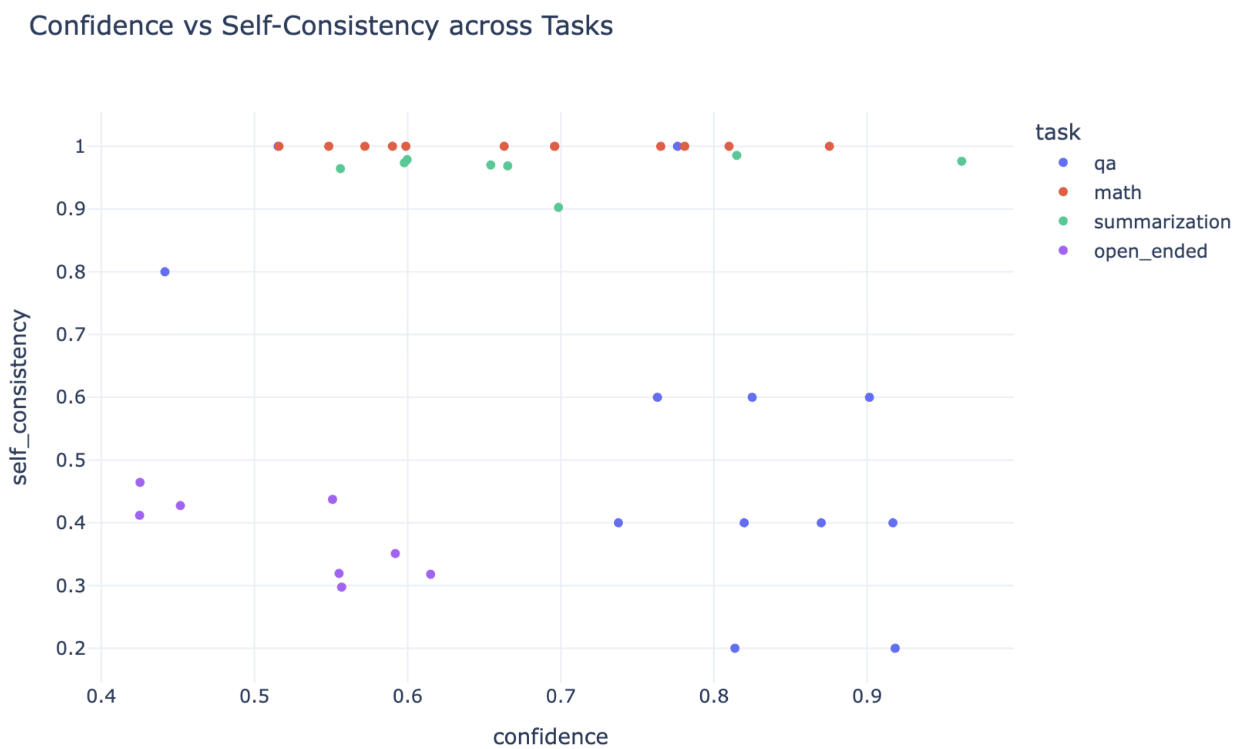

Figure 5. Confidence versus self consistency across tasks.
This plot compares token level confidence with disagreement based uncertainty estimation. Higher consistency across sampled generations generally corresponds to lower uncertainty.

5.4 Calibration Analysis

Calibration analysis revealed that the language model was not perfectly calibrated across tasks. Reliability diagrams showed that the model frequently exhibited overconfidence, particularly in factual hallucination scenarios and long form generation tasks. In several confidence intervals, empirical accuracy fell below predicted confidence levels, indicating that the model assigned excessive confidence to incorrect outputs.

Calibration quality was strongest for factual QA tasks, where shorter deterministic answers allowed uncertainty estimates to align more closely with correctness. Mathematical reasoning tasks displayed more unstable calibration behavior because intermediate reasoning uncertainty was not always reflected in final answer confidence.

Summarization and open ended generation tasks further demonstrated the difficulty of evaluating calibration in generative settings. Since multiple valid outputs may exist, confidence estimates do not always correspond directly to conventional correctness metrics. Nevertheless, reliability analysis still provided useful insight into the general relationship between model confidence and observed performance.

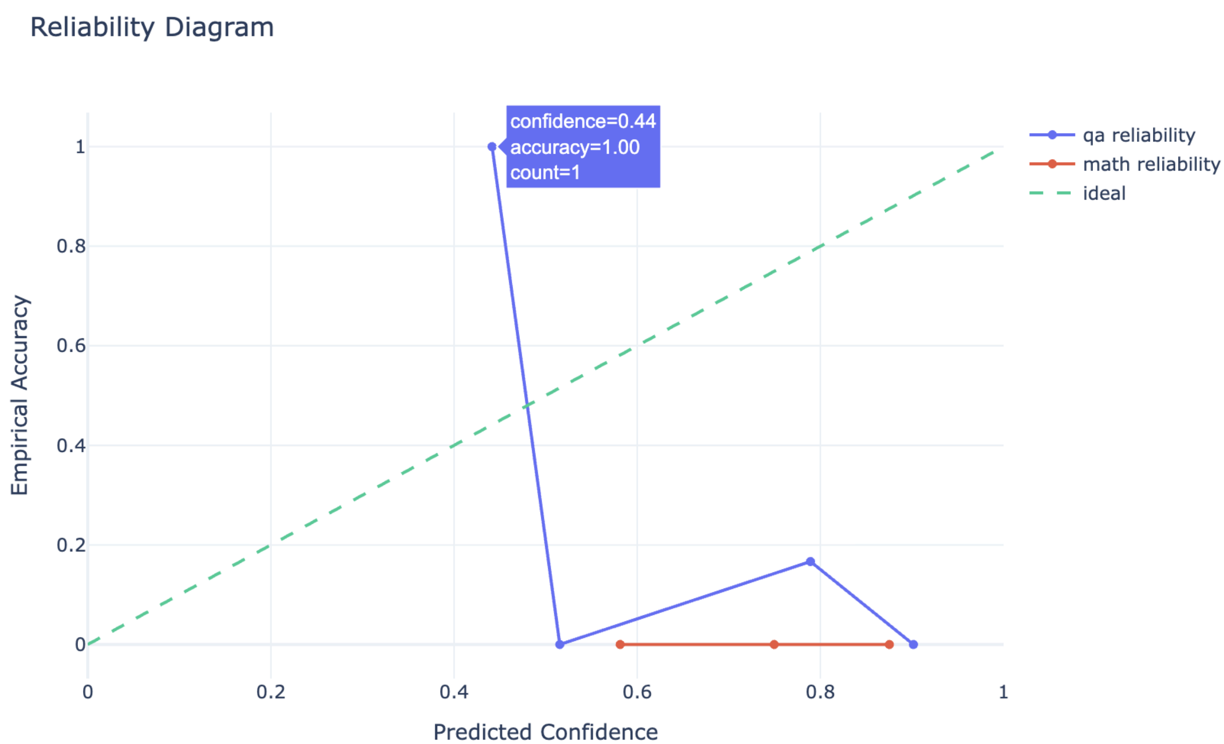

Figure 6. Reliability diagram for factual QA and mathematical reasoning tasks.
The reliability diagram compares predicted confidence with empirical accuracy. Deviations below the diagonal indicate overconfidence in model predictions.

5.5 Task Level Metric Analysis

Task level summary metrics further illustrate the relationship between uncertainty, confidence, and model performance. Factual QA tasks achieved the highest confidence and lowest entropy values, while open ended generation exhibited the largest uncertainty variability. Mathematical reasoning tasks demonstrated moderate confidence but higher inconsistency across sampled generations due to reasoning complexity.

The heatmap visualization provides a compact overview of average confidence, entropy, self consistency, and task specific performance metrics across all evaluated task categories. Together, these analyses highlight that uncertainty estimation behavior is highly task dependent and that no single uncertainty metric performs optimally across all generation settings.

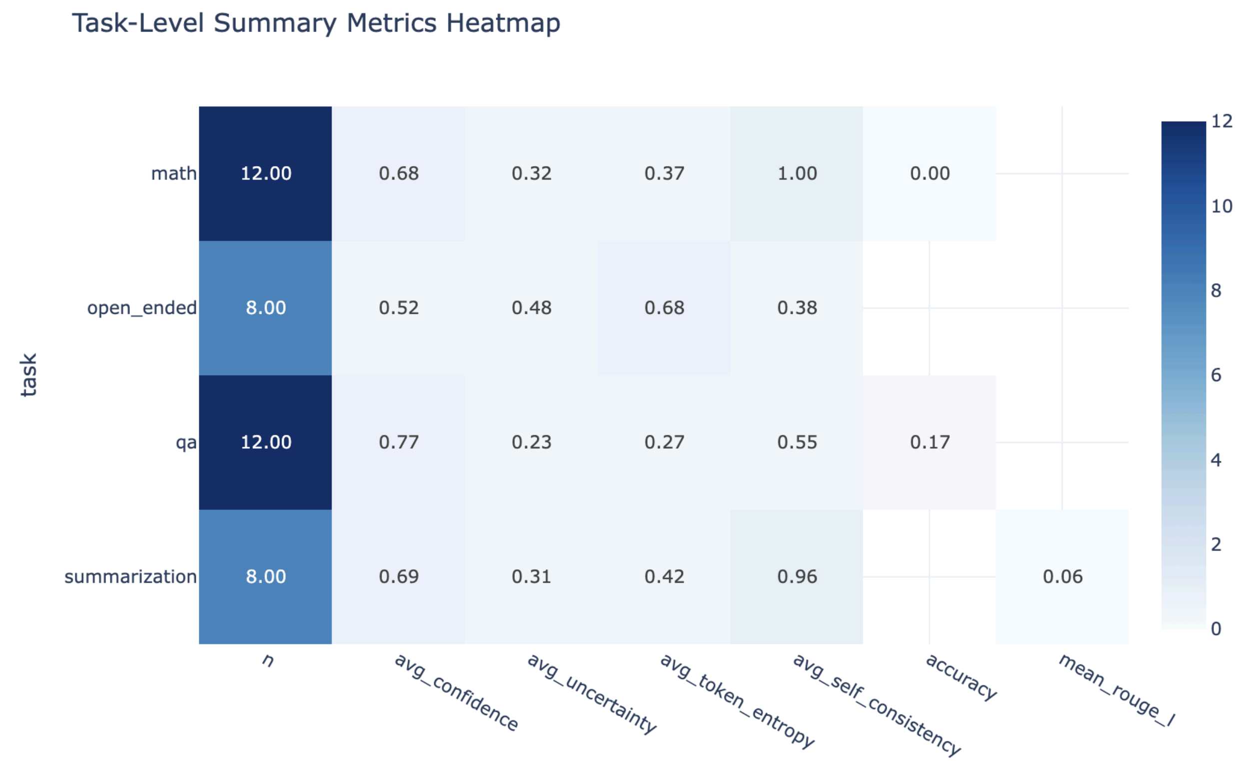

Figure 7. Task level summary metrics heatmap.
This visualization summarizes confidence, entropy, self consistency, and performance related statistics across all evaluated task categories.

6. Discussion

The experimental results demonstrate that uncertainty estimation methods can provide meaningful signals about the reliability of autoregressive language model outputs. Across multiple task categories, lower uncertainty was generally associated with more accurate and consistent predictions, while higher uncertainty frequently corresponded to hallucinations, unstable reasoning, or semantically inconsistent generations. Although the implemented methods are relatively lightweight compared to large scale Bayesian or ensemble approaches, they were still able to capture useful patterns regarding model confidence and reliability.

Among the evaluated methods, token entropy emerged as the most practical uncertainty estimation technique. Since entropy can be computed directly from token probability distributions during generation, it introduces minimal computational overhead and integrates naturally into existing autoregressive inference pipelines. Entropy based uncertainty estimation performed particularly well for factual question answering tasks where answers were relatively deterministic and generation variability was limited. However, entropy alone was less effective for capturing higher level semantic uncertainty, especially in tasks where multiple plausible outputs existed.

Average sequence log probability also provided useful confidence estimates and correlated reasonably well with overall generation quality. Nevertheless, the experiments showed that fluent language generation does not necessarily imply factual correctness. In several cases, the model generated incorrect responses with relatively high sequence confidence. This highlights an important limitation of probability based uncertainty estimation methods in generative language modeling. High likelihood under the model distribution may reflect linguistic fluency rather than true factual reliability.

Self consistency based uncertainty estimation produced the strongest qualitative uncertainty signals across reasoning intensive tasks. When multiple stochastic generations converged toward similar outputs, the model was generally more reliable. Conversely, disagreement among sampled generations often corresponded to uncertainty in reasoning or factual inconsistencies. This behavior was especially evident in mathematical reasoning and open ended generation tasks where generation variability was naturally higher. However, self consistency methods require multiple generation passes for each prompt, leading to substantially higher inference cost and runtime compared to entropy based approaches.

The experiments also revealed that uncertainty behavior is highly task dependent. Factual QA tasks exhibited relatively stable confidence distributions and lower entropy values because answers were short and deterministic. Mathematical reasoning tasks demonstrated more complex uncertainty patterns, with uncertainty often fluctuating throughout intermediate reasoning steps even when final answers were correct. Summarization and open ended generation tasks produced significantly broader uncertainty distributions due to increased semantic variability and the existence of multiple valid outputs.

Calibration analysis further highlighted important limitations of current language models. The reliability diagrams showed that the model frequently exhibited overconfidence, particularly in hallucination prone settings and long form generation tasks. This observation is consistent with recent research showing that modern language models often produce highly fluent outputs even when underlying factual certainty is weak. Although uncertainty estimates correlated with correctness overall, calibration remained imperfect and high confidence failures were still observed.

Several limitations of this work should also be acknowledged. Due to computational constraints, experiments were conducted on representative subsets of benchmark datasets rather than full evaluation splits. Additionally, the project focused on relatively lightweight uncertainty estimation methods that could be implemented efficiently within a limited runtime environment. More advanced approaches such as Bayesian neural networks, ensemble based uncertainty estimation, or semantic entropy clustering were beyond the scope of this implementation. The evaluation metrics themselves also have limitations, particularly for open ended generation tasks where correctness is inherently subjective.

Despite these limitations, the project demonstrates that practical uncertainty estimation can provide valuable information about language model reliability without requiring major architectural modifications or expensive retraining procedures. Lightweight uncertainty estimation methods such as token entropy and self consistency analysis can therefore serve as useful components in trustworthy AI systems, particularly in settings where detecting unreliable or hallucinated outputs is important.

7. Future Work

There are several promising directions for extending this work and improving uncertainty estimation in autoregressive language models. One natural extension would be to explore larger and more capable language models in order to study how uncertainty behavior changes with increasing model scale. Larger instruction tuned models may exhibit stronger calibration properties and more reliable uncertainty estimates, particularly for reasoning intensive tasks such as mathematical problem solving and factual question answering.

Another important direction involves semantic uncertainty estimation. The methods implemented in this project primarily rely on token probability distributions and response disagreement, which do not always capture deeper semantic equivalence between generated outputs. Future work could incorporate semantic entropy based methods that cluster responses according to meaning rather than surface level token variation. Such approaches may provide more accurate uncertainty estimates for summarization and open ended generation tasks where multiple semantically valid outputs exist.

Bayesian and ensemble based approaches also remain promising areas for future investigation. Techniques such as Monte Carlo dropout, model ensembles, or Bayesian transformer approximations could potentially provide stronger epistemic uncertainty estimates compared to single model probability based methods. However, these methods typically introduce substantial computational overhead and would require careful study of the trade off between uncertainty quality and inference efficiency.

Future research could also explore uncertainty aware decoding strategies. Instead of generating responses independently from uncertainty estimation, the decoding process itself could dynamically adapt based on predicted uncertainty levels. For example, the model could abstain from answering, request clarification, or retrieve external information whenever uncertainty exceeds a predefined threshold. Such mechanisms could significantly improve the safety and trustworthiness of deployed language model systems.

Another promising direction involves retrieval augmented generation and external knowledge integration. Many hallucination related failures arise because autoregressive language models rely solely on internal parametric memory during generation. Incorporating retrieval systems or external knowledge bases may improve both factual accuracy and uncertainty calibration by grounding generations in verifiable information.

The current project also evaluated uncertainty using relatively small dataset subsets due to runtime and hardware constraints. Future work could extend the evaluation framework to larger benchmark datasets and more diverse task settings. Additional evaluation domains such as code generation, multilingual reasoning, scientific question answering, and conversational dialogue systems may provide further insight into the strengths and limitations of current uncertainty estimation methods.

Finally, human centered evaluation remains an important area for future exploration. Although automatic metrics such as entropy, calibration curves, and exact match accuracy provide useful quantitative analysis, uncertainty estimation ultimately affects how humans interact with AI systems. Studying how users interpret and respond to uncertainty signals could help design more transparent and trustworthy human AI interaction frameworks.

8. Conclusion

This project investigated practical uncertainty quantification techniques for autoregressive language models across multiple natural language generation tasks. The study focused on estimating uncertainty at both token and sentence levels using lightweight methods that can be integrated directly into standard transformer based generation pipelines. Experiments were conducted across factual question answering, mathematical reasoning, summarization, and open ended generation tasks in order to analyze how uncertainty behaves under different generation settings.

The results demonstrate that uncertainty estimates correlate meaningfully with model reliability and prediction correctness across several tasks. Token entropy and sequence level confidence provided useful low cost uncertainty signals, while self consistency based methods produced stronger semantic uncertainty estimates for reasoning intensive tasks. At the same time, the experiments also revealed important limitations of current language models, particularly regarding calibration and overconfidence in hallucinated or incorrect outputs.

One of the central observations from this work is that uncertainty behavior is highly task dependent. Deterministic tasks such as factual question answering exhibited relatively stable confidence distributions, whereas summarization and open ended generation showed substantially higher uncertainty variability due to semantic diversity and the existence of multiple plausible outputs. Mathematical reasoning tasks further highlighted that uncertainty can fluctuate significantly during intermediate reasoning steps even when final answers are correct.

The project also demonstrated that practical uncertainty estimation can be implemented efficiently using open source models and standard generation pipelines without requiring expensive retraining or architectural modifications. Lightweight uncertainty estimation methods therefore represent a promising direction for improving the transparency, interpretability, and trustworthiness of generative AI systems.

Although the implemented methods remain imperfect and several limitations persist, the experiments provide useful insight into the strengths and weaknesses of current uncertainty estimation approaches. As language models continue to be integrated into increasingly important real world applications, reliable uncertainty quantification will remain a critical component of trustworthy and safe AI deployment.

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